US Patent & Trademark Office
Patent Public Search | Text View

United States Patent 12384356
Kind Code B2
Date of Patent August 12, 2025
Inventor(s) Christen; Erik J. et al.

Electrified vehicle control during trailer towing

Abstract

A system and method for a vehicle having a traction battery, an electric machine, and a secondary power source, such as an engine, operate the electric machine and secondary power source to sustain a state of charge (SOC) of the traction battery when the SOC is either: a) below a first threshold and above a second threshold; or b) below a third threshold; and to deplete the SOC of the traction battery when the SOC is between the second and the third thresholds in response to detecting an operating condition associated with high Amp-hour (Ah) throughput of the traction battery, such as towing a trailer. The system and method operate the electric machine and the secondary power source to deplete the SOC when the SOC is above the third threshold and sustain the SOC when the SOC is below the third threshold when the condition is not detected.

Inventors: **Christen; Erik J. (Royal Oak, MI), Kucab; Gregory (Ypsilanti, MI)**

Applicant: **FORD GLOBAL TECHNOLOGIES, LLC** (Dearborn, MI)

Family ID: **85775331**

Assignee: **Ford Global Technologies, LLC (Dearborn, MI)**

Appl. No.: **17/493495**

Filed: **October 04, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20230104277 A1	Apr. 06, 2023

Publication Classification

Int. Cl.: **B60W20/13** (20160101); **B60D1/62** (20060101); **B60W50/14** (20200101)

U.S. Cl.:

CPC	**B60W20/13** (20160101); **B60W50/14** (20130101); B60D1/62 (20130101); B60W2420/403 (20130101); B60W2510/244 (20130101); B60W2520/105 (20130101); B60W2520/125 (20130101); B60W2530/203 (20200201); B60W2540/215 (20200201)

Field of Classification Search

CPC: B60W (20/13); B60W (2530/203)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10384667	12/2018	Frank et al.	N/A	N/A
10773602	12/2019	Caliskan et al.	N/A	N/A
2015/0291145	12/2014	Yu	180/65.23	B60W 20/13
2017/0001581	12/2016	Rinker	N/A	F02D 41/021
2019/0225154	12/2018	Ionascu	N/A	G06F 18/256
2021/0086628	12/2020	Kava et al.	N/A	N/A
2022/0041156	12/2021	Imamura	N/A	B60K 6/445

Primary Examiner: Mustafa; Imran K

Attorney, Agent or Firm: Brooks Kushman PC

Background/Summary

TECHNICAL FIELD
(1) This disclosure relates to control of an electrified vehicle while towing a trailer.
BACKGROUND
(2) As more and more vehicles are electrified, high voltage (HV) battery packs are increasing in capacity and number of battery cells to provide expected travel range and vehicle performance. An HV battery pack may include a controller, such as a battery electrical control module (BECM), several cell monitor and management (CMM) modules, and various other controllers and electronics to monitor and control charging and operation of individual battery cells that are connected together to provide the HV battery pack output voltage and current. Various conditions during charging and operation may impact HV battery pack performance and health. One metric used to monitor or measure the impact of battery utilization on battery health is Amp-hour (Ah) throughput. Higher Ah throughput associated with extended use in unfavorable operating conditions or high-demand applications such as towing a trailer may be associated with more rapid deterioration of the battery State of Health (SoH).
SUMMARY
(3) Embodiments of the disclosure include an electrified vehicle comprising a traction battery, an electric machine powered by the traction battery to selectively provide propulsion to the vehicle, an engine coupled to the electric machine, and a controller programmed to, in response to detecting a trailer connected to the electrified vehicle, operate the electric machine and the engine to sustain a state of charge (SOC) of the traction battery when the SOC is either between a first threshold and a second threshold, or below a third threshold, and operate the electric machine and the engine to

deplete the SOC when the SOC is either above the first threshold or between the second threshold and the third threshold. The first threshold is higher than the second threshold and the second threshold is higher than the third threshold. In one embodiment, the first threshold corresponds to 70% SOC and the second threshold corresponds to 30% SOC.
(4) The electrified vehicle may include at least one sensor mounted on the vehicle and configured to provide a signal to the controller to detect the trailer. The at least one sensor may include a camera. The controller may be programmed to detect the trailer when a trailer electrical circuit configured to power trailer lights is connected to a corresponding electrical circuit of the electrified vehicle. The controller may be programmed to detect the trailer by comparing a measured vehicle acceleration to an expected vehicle acceleration associated with a driver demand torque. The electrified vehicle may include a human-machine interface (HMI), wherein the controller is programmed to detect the trailer in response to input received by the HMI and communicated to the controller.
(5) Embodiments may also include a method for controlling an electrified vehicle having a traction battery coupled to an electric machine to selectively provide propulsive torque to the vehicle, and a secondary power source configured to supply power to at least one of the electric machine and the traction battery. The method includes, by a vehicle controller, controlling the electric machine and the secondary power source to sustain a state of charge (SOC) of the traction battery when the SOC is either below a first threshold and above a second threshold or below a third threshold in response to detecting an operating condition associated with traction battery amp-hour throughput exceeding an associated throughput threshold; and controlling the electric machine and the secondary power source to deplete the SOC of the traction battery when the SOC is above the third threshold, and to sustain the SOC when the SOC is below the third threshold when the operating condition is not detected. The method may also include controlling the electric machine and the secondary power source to deplete the SOC of the traction battery when the SOC is either above the first threshold, or below the second threshold and above the third threshold in response to detecting the operating condition.
(6) The secondary power source may include an internal combustion engine. The operating condition may include towing a trailer. The operating condition may be detected in response to receiving input from a human-machine interface (HMI) of the electrified vehicle, and/or detecting the completion of an electrical circuit between the trailer and the electrified vehicle. Detection of the operating condition may be based on comparing measured acceleration of the electrified vehicle for a specified torque request to an associated expected acceleration.
(7) Various embodiments may include a system for an electrified vehicle having an electric machine powered by a traction battery and an engine that includes a controller programmed to, in response to detecting an operating condition associated with traction battery amp-hour throughput exceeding an associated throughput threshold: sustain a state of charge (SOC) of the traction battery when the SOC is either: a) below a first threshold and above a second threshold; or b) below a third threshold; and deplete the SOC of the traction battery when the SOC is between the second and the third thresholds. The controller may be further programmed to detect the operating condition in response to detecting a trailer coupled to the electrified vehicle. The controller may be further programmed to, in response to not detecting the operating condition: control vehicle operation to deplete the SOC when the SOC is above the third threshold; and to sustain the SOC when the SOC is below the third threshold. The controller may be further programmed to detect the operating condition based on input from a human-machine interface (HMI) of the electrified vehicle. The controller may be further programmed to detect the operating condition based on an electrical connection between the trailer and the electrified vehicle.
(8) Embodiments of the disclosure may provide one or more associated advantages. For example, inserting a charge sustaining zone within a typical charge depletion zone when towing a trailer or under other high demand conditions associated with high Ah throughput reduces the Ah throughput

over the lifetime of the vehicle and mitigates the impact of high throughput on HV battery SoH. For many existing vehicles, the control strategy may be implemented via a software update to leverage existing vehicle hardware to provide a relatively low cost implementation. This solution may be customer selectable via the HMI and may improve customer satisfaction and vehicle residual value by maintaining battery SoH.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates an example of an electrified vehicle in a high demand use scenario such as towing a trailer and having a control strategy to mitigate the effect on HV battery SoH.

(2) FIG. **2** is a block diagram illustrating a representative electrified vehicle implemented as a plug-in hybrid electric vehicle (PHEV) having a controller to control powertrain operation during trailer towing or other high demand use scenarios according to one or more embodiments.

(3) FIG. **3** is a graphical representation of electrified vehicle operation according to a prior art control strategy.

(4) FIG. **4** illustrates operating zones for a representative electrified vehicle with a battery health zone to reduce Ah throughput of the HV battery while towing a trailer.

(5) FIG. **5**A illustrates operation of an electrified vehicle having a battery SOC above an associated battery health zone upper SOC threshold at the beginning of a trip.

(6) FIG. **5**B illustrates operation of an electrified vehicle having a battery SOC between upper and lower SOC thresholds for an associated battery health zone at the beginning of a trip.

(7) FIG. **5**C illustrates operation of an electrified vehicle having a battery SOC below a lower SOC threshold of an associated health zone SOC at the beginning of a trip

(8) FIG. **6** is a graph illustrating operation of a system or method for controlling an electrified vehicle during trailer towing to maintain operation within a battery health zone.

DETAILED DESCRIPTION

(9) Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments can take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures can be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

(10) The present inventors have recognized that particular use scenarios of an electrified vehicle may adversely affect the HV battery SoH. For example, an electrified vehicle towing a trailer for a significant portion of its trips may experience more rapid HV battery degradation relative to a vehicle that has a lower percentage of trips with conditions that quickly discharge the HV battery. Furthermore, that Ah throughput of the HV battery is one metric that may be used to monitor or evaluate the HV battery SoH. As such, reducing Ah throughput of the HV battery over the lifetime of the vehicle, specifically during trailing towing events, may improve HV battery performance and durability over the lifetime of the vehicle. As such, various embodiments according to the disclosure identify events that may otherwise result in higher impact to the HV battery SoH and control the vehicle to reduce or eliminate the impact. In one or more embodiments, the electrified

vehicle is controlled to operating within a battery health zone having upper and lower battery SOC thresholds for operating in a charge sustaining mode, which reduces operation in a charge depletion mode during a detected use scenario, such as trailer towing. Under these use scenarios, a vehicle starting a trip with the HV battery SOC above the health zone will operate in charge depletion mode until the SOC crosses into the health zone, and then operate in charge sustaining mode. A vehicle starting a trip with the HV battery SOC within the health zone will operate in charge sustaining mode. A vehicle starting a trip with the HV battery SOC below the lower threshold of the health zone will operate the same as when the use scenario has not been detected, according to normal mode charge depletion and charge sustaining SOC thresholds. Stated differently, the control strategy inserts a charge sustaining battery health zone for trailer towing in the middle of the normal charge depletion zone, resulting in two charge depletion zones and two charge sustaining zones as compared to normal operation with only one charge depletion zone and one charge sustaining zone.

(11) FIG. **1** illustrates an electrified vehicle **112** towing a trailer **110**. Trailer towing is a representative high demand use scenario that may initiate an HV battery SoH protection or mitigation control strategy according to this disclosure. Vehicle **112** may have more than one control mode or strategy to reduce or eliminate the impact of high Ah throughput on long term battery SoH with the particular strategy selected based on satisfying the corresponding entrance conditions. As such, those of ordinary skill in the art will recognize that any embodiments described with respect to a trailer towing scenario can similarly be applied to other high demand use scenarios, which may have different upper and lower battery health zone thresholds, different entry conditions, etc. Vehicle **112** may detect towing of a trailer **110** based on one or more inputs from a vehicle operator and various sensors. For example, an operator may select a tow/haul mode using a physical switch or other input via a human-machine interface (HMI) **100** (FIG. **2**). Vehicle **112** may include one or more cameras, such as a rear view or backup camera **80** and side view camera **82** integrated with the side view mirror(s) with associated image processing to detect trailer **110**. Other sensors, such as a trailer hitch sensor **90**, electronic suspension sensor (not shown), trailer electrical connection, etc. may also be used. The vehicle system controller **148** (FIG. **2**) may also include one or more algorithms to detect trailer **110** based on grade, torque request, known vehicle properties such as weight, acceleration, drag, etc.

(12) FIG. **2** depicts a representative embodiment of an electrified vehicle with at least one controller to control the vehicle during trailer towing or other high demand use scenarios according to the disclosure. In the representative vehicle embodiment of FIG. **2**, vehicle **112** is an electrified vehicle, such as a plug-in hybrid electric vehicle (HEV) in this example, but may also be a fully electrified battery electric vehicle (BEV) or other type of vehicle depending on the particular implementation. As explained in greater detail below, various types of electrified vehicles may include more than one power source such that a HV battery may be operated in either charge depletion or charge sustaining mode or to switch to an alternative power source to maintain the battery SOC within a predetermined health zone for a particular high demand use scenario. Vehicle **112** may comprise one or more electric machines **114** mechanically connected to a transmission **116**. The electric machines **114** may be capable of operating as a motor or a generator. In addition, the transmission **116** is mechanically connected to an internal combustion engine **118**. The transmission **116** is also mechanically connected to a drive shaft **20** that is mechanically connected to the wheels **122**. The electric machines **114** can provide propulsion and regenerative braking capability when the engine **118** is turned on or off. During regenerative braking, the electric machines **114** act as generators and can provide fuel economy benefits by recovering energy that would normally be lost as heat in the friction braking system. The electric machines **114** may also reduce vehicle emissions by allowing the engine **118** to operate at more efficient speeds and allowing the hybrid-electric vehicle **112** to be operated in electric mode with the engine **118** off under certain conditions.

(13) An HV traction battery or battery pack **124** stores energy that can be used by the electric machines **114** in a plurality of low voltage cells connected in groups or strings to provide a desired energy storage capacity and output voltage/current. Vehicle **112** may include more than one battery pack that may be used simultaneously or in sequence. For example, vehicle **112** may include a second battery pack (not shown) that provides an emergency reserve to extend the electric range, power external equipment, and/or to provide additional acceleration for applications such as trailer towing.
(14) Vehicle battery pack **124** typically provides a high voltage DC output. Each group or string of cells may have an associated cell monitoring module (CMM) **160** that measures voltages for individual battery cells or groups of cells and controls various battery functions, such as cell balancing, for example. Each CMM **160** may communicate with a battery controller, sometimes referred to as a battery energy control module (BECM) **180**. The BECM may communicate with one or more other vehicle controllers such as vehicle system controller **148** over a wired or wireless vehicle network to provide higher-level control functions of the traction battery **124** as described herein.
(15) The traction battery **124** is electrically connected to one or more power electronics modules. One or more contactors **142** may isolate the traction battery **124** from other components when opened and connect the traction battery **124** to other components when closed. A power electronics module **126** is also electrically connected to the electric machines **114** and provides the ability to bi-directionally transfer energy between the traction battery **124** and the electric machines **114**. For example, a typical traction battery **124** may provide a DC voltage while the electric machines **114** may require a three-phase AC current to function. The power electronics module **126** may convert the DC voltage to a three-phase AC current as required by the electric machines **114**. In a regenerative mode, the power electronics module **126** may convert the three-phase AC current from the electric machines **114** acting as generators to the DC voltage required by the traction battery **124**. The description herein is equally applicable to an electrified vehicle implemented as a pure electric vehicle, often referred to as a battery electric vehicle (BEV). For a BEV, the hybrid transmission **116** may be a gear box connected to an electric machine **114** and the engine **118** may be omitted.
(16) In addition to providing energy for propulsion during a charge depletion mode, the traction battery **124** may provide energy for other vehicle electrical systems. A typical system may include a DC/DC converter module **128** that converts the high voltage DC output of the traction battery **124** to a low voltage DC supply that is compatible with other vehicle loads. Other high-voltage loads, such as compressors and electric heaters, may be connected directly to the high-voltage without the use of a DC/DC converter module **128**. The low-voltage systems may be electrically connected to an auxiliary battery **130** (e.g., 12V, 24V, or 48V battery).
(17) The electrified vehicle **112** may be a BEV or a plug-in hybrid vehicle in which the traction battery **124** may be recharged by an external power source **136**, or a standard hybrid that charges traction battery **124** from operating electric machines as a generator during a charge sustaining mode but does not receive power from an external power source. The external power source **136** may be a connection to an electrical outlet. The external power source **136** may be electrically connected to electric vehicle supply equipment (EVSE) **138**. The EVSE **138** may provide circuitry and controls to regulate and manage the transfer of energy between the power source **136** and the vehicle **112**. In other embodiments, the vehicle **112** may employ wireless charging, which may be referred to as hands-free or contactless charging that uses inductive or similar wireless power transfer.
(18) The external power source **136** may provide DC or AC electric power to the EVSE **138**. The EVSE **138** may have a charge connector **140** for plugging into a charge port **134** of the vehicle **112**. The charge port **134** may be any type of port configured to transfer power from the EVSE **138** to the vehicle **112**. The charge port **134** may be electrically connected to an on-board power

conversion module **132** having an associated battery charger controller. The power conversion module **132** may condition the power supplied from the EVSE **138** to provide the proper voltage and current levels to the traction battery **124**. The power conversion module **132** may interface with the EVSE **138** to coordinate the delivery of power to the vehicle **112**. The EVSE connector **140** may have pins that mate with corresponding recesses of the charge port **134**. Alternatively, various components described as being electrically connected may transfer power using a wireless inductive coupling as previously described.

(19) One or more wheel brakes **144** may be provided for friction braking of the vehicle **112** and preventing motion of the vehicle **112**. The wheel brakes **144** may be hydraulically actuated, electrically actuated, or some combination thereof. The wheel brakes **144** may be a part of a brake system **150**. The brake system **150** may include other components that are required to operate the wheel brakes **144**. For simplicity, the figure depicts a single connection between the brake system **150** and one of the wheel brakes **144**. A connection between the brake system **150** and the other wheel brakes **144** is implied.

(20) One or more electrical loads **146** may be connected to the high-voltage bus. The electrical loads **146** may have an associated controller that operates the electrical load **146** when appropriate. Examples of electrical loads **146** may be a heating module or an air-conditioning module.

(21) The various components described may have one or more associated controllers to control and monitor the operation of the components. It should be understood that any one of the representative controllers can collectively be referred to as a "controller" that controls various actuators in response to signals from various sensors to control the vehicle. The controllers may communicate via a vehicle network that may be implemented as a serial bus (e.g., Controller Area Network (CAN)) or via discrete conductors. In addition, system controller **148** may be present to coordinate the operation of the various components and may communicate directly or indirectly with one or more other vehicle controllers, such as BECM **180**, a body controller or control module, and a battery charger controller or control module. HMI **100** may communicate with system controller **148** to receive input from a vehicle operator and provide visual, audio, and/or haptic output to vehicle occupants. HMI **100** may also provide coordination of various vehicle telematics and infotainment system functions any may communicate via a modem with external services via a cellular or satellite modem, for example. HMI **100** may prompt for input to confirm trailer towing mode or other high demand mode operation. In some applications HMI **100** may allow customization of various entry conditions or operational limits for a battery health zone to allow users to adjust vehicle performance for a particular trip or time period within limits set by the system.

(22) Each controller, such as vehicle system controller **148**, may include a microprocessor or central processing unit (CPU) in communication with various types of memory or non-transitory computer readable storage devices or media. Computer readable storage devices or media may include volatile and nonvolatile or persistent storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the processor is powered down. Computer-readable storage devices or media may be implemented using any of a number of known memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or any other electric, magnetic, optical, solid state, or combination memory devices capable of storing data, some of which represent executable instructions, used by the controller to implement various algorithms or control strategies to control the vehicle **112** via various vehicle components or subsystems.

(23) Control logic, functions, code, software, strategy etc. performed by one or more processors or controllers may be represented by block diagrams or flow charts (as in FIG. **6**, for example), or similar diagrams in one or more figures (as in FIGS. **1-5**, for example). These figures provide representative control strategies, algorithms, and/or logic for operation of a system or method

according to the disclosure that may be implemented using one or more processing strategies such as event-driven, interrupt-driven, multi-tasking, multi-threading, and the like. As such, various steps or functions illustrated or described may be performed in the sequence as illustrated or described, in parallel, or in some cases omitted. Although not always explicitly illustrated or described, one of ordinary skill in the art will recognize that one or more of the steps or functions may be repeatedly performed depending upon the particular processing strategy being used. Similarly, the order of processing is not necessarily required to achieve the features and advantages described herein, but is provided for case of illustration and description. The control logic may be implemented primarily in software executed by a microprocessor-based vehicle, engine, and/or powertrain controller, such as system controller **148**. Of course, the control logic may be implemented in software, hardware, or a combination of software and hardware in one or more controllers depending upon the particular application. When implemented in software, the control logic may be provided in one or more non-transitory computer-readable storage devices or media having stored data representing code or instructions executed by a computer to control the vehicle or its subsystems as previously described.

(24) FIG. **3** is a graphical representation **300** of electrified vehicle operation according to a prior art control strategy illustrating HV battery % SOC as a function of time over representative driving cycles, which may include city and/or highway driving. The control strategy illustrated includes a charge depletion zone **302** and a charge sustaining (CS) zone **304** defined by an associated threshold **306**, which may be 17% SOC, for example. During operation beginning with a complete charge indicated by SOC_MAX, the vehicle is operated in a charge depleting (CD) zone or mode **302** and the measured or estimated SOC **310** continues to decrease at different rates over various charge depleting cycles **314**, which may include city and/or highway cycles, depending on the torque demand of the electric machine. When SOC crosses threshold **306**, the control strategy switches to a charge sustaining mode or zone **304** with the electric machine and engine operated to maintain a target SOC **308** within the charge sustaining zone **304**. While in this mode, the SOC may increase and decrease around the target value to provide useable electric power to the electric machine as indicated at **312** to keep the SOC above the SOC_LOW threshold.

(25) As also illustrated in the graphical representation **300**, battery power **320** may be limited between a maximum charging power **322** and a maximum discharging power **324**. The maximum charging power increases from zero at SOC_MAX to its limit at SOC_HIGH. The maximum discharging power ranges from zero at SOC_MIN to its limit at SOC_LOW. As illustrated by the power axis **320**, battery operation includes some charge capacity above SOC_MAX and below SOC_MIN but is limited by the controller to provide sufficient operational margins and avoid damage to the battery by overcharging or over discharging.

(26) FIG. **4** illustrates operating zones or modes for a representative electrified vehicle with a battery health mode or zone **410** with a control strategy to reduce Ah throughput of the HV battery while towing a trailer or operating under similar high-demand scenarios. Health zone **410** is defined by a first SOC threshold **412** and a second SOC threshold **414**. After detecting a trailer coupled to the electrified vehicle, the controller operates the electric machine and secondary power source, such as an internal combustion engine, in charge sustaining mode to maintain a target SOC while the battery SOC is within the Health zone **410**, i.e. when the SOC is between the first threshold **412** and second threshold **414**. The controller also operates the powertrain in charge sustaining mode when the SOC falls below a third threshold **416**, which corresponds to the normal CS operation threshold when not towing a trailer. As illustrated, the first threshold **412** is higher than the second threshold **414**, which in turn is higher than the third threshold **416**. In one embodiment, the health zone is defined by a first threshold **412** at 70% SOC and a second threshold **414** at 30% SOC. The third threshold **416** for normal CS mode is at 17% SOC. However, the programmable thresholds may vary depending on the particular application and implementation.

(27) As illustrated in FIG. **4**, health zone **410** effectively creates alternating charge depletion and

charge sustaining modes or zones including a first charge depletion zone **420** above the first threshold **412**, a first charge sustaining zone **410** between the first threshold **412** and the second threshold **414**, a second charge depletion zone **422** between the second threshold **414** and the third threshold **416**, and a second charge sustaining zone **424** below the third threshold **416**. As such, the control strategy according to various embodiments of the disclosure includes two charge sustaining zones and two charge depletion zones in contrast with the prior art control strategy illustrated in FIG. **3** with only a single CD zone and a single CS zone.

(28) FIG. **5**A illustrates operation of an electrified vehicle having a battery SOC **500**A above an associated battery health zone upper SOC threshold **412** at the beginning of a trip after detecting a trailer or similar high demand scenario. The control strategy operates the powertrain in charge depletion zone **420** until the SOC **500**A crosses the health zone upper SOC threshold **412**, and then operates in charge sustaining zone **410** with a target SOC **510**A. In this region, the HV battery may still provide additional discharge power to the electric machine, which is beneficial during trailer tow operation (i.e. to provide better acceleration).

(29) FIG. **5**B illustrates operation of an electrified vehicle having a battery SOC **500**B between upper health zone threshold **412** and lower health zone threshold **414** of an associated battery health zone at the beginning of a trip after detecting a trailer or similar high demand scenario. As illustrated, the SOC **500**B starts within health zone **410**. As such, the controller operates the powertrain in charge sustaining mode with a target SOC **510**B. Again, in this region, the HV battery may still provide additional discharge power to the electric machine, which is beneficial during trailer tow operation (i.e. to provide better acceleration).

(30) FIG. **5**C illustrates operation of an electrified vehicle having a battery SOC **500**C below a lower health zone threshold **414** at the beginning of a trip after detecting a trailer or similar high demand scenario. The control strategy operates the powertrain in charge depletion zone **422** until the SOC **500**C falls below threshold **416**, and then operates the powertrain in charge sustaining zone **424** with a target SOC **510**C to maintain the SOC above the SOC_LOW threshold. In this region little or no additional discharge power is available for the electric machine to provide better acceleration.

(31) As shown in FIGS. **2**, **4**, and **5**A-**5**C, electrified vehicle **112** includes one or more controllers **148**, **180** programmed to, in response to detecting a trailer **110** connected to electrified vehicle **112**, operate the electric machine(s) **114** and the engine **118** to sustain SOC of the traction battery **124** when the SOC is either between a first threshold **412** and a second threshold **414** or below a third threshold **416**, and operate the electric machine(s) **114** and the engine **118** to deplete the SOC when the SOC is either above the first threshold **114** or between the second threshold **414** and the third threshold **416**. Similarly, a method for controlling electrified vehicle **112** including traction battery **124** coupled to an electric machine **114** to selectively provide propulsive torque to the vehicle **112**, and a secondary power source **118** configured to supply power to at least one of the electric machine **114** and the traction battery **124**, includes, by a vehicle controller **148**, **180**: controlling the electric machine **114** and the secondary power source **118** to sustain a state of charge (SOC) of the traction battery **124** when the SOC is either below a first threshold **412** and above a second threshold **414** or below a third threshold **416** in response to detecting an operating condition associated with traction battery amp-hour throughput exceeding an associated throughput threshold. The method also includes controlling the electric machine **114** and the secondary power source **118** to deplete the SOC of the traction battery **124** when the SOC is above the third threshold **416**, and to sustain the SOC when the SOC is below the third threshold **416** when the operating condition is not detected.

(32) FIG. **6** is a flowchart illustrating operation of a system or method **600** for controlling an electrified vehicle during trailer towing to maintain battery SOC within a battery health zone. HV battery SOC is monitored by one or more controllers **148**, **180** as represented at **610**. The controller detects whether a trailer is connected to the electrified vehicle (or conditions related to various

other high demand use scenarios) as represented at **620**. A trailer may be detected in response to a user entered input to select a tow/haul mode via an HMI as represented at **622**. Alternatively, or in combination, a camera, radar, lidar, or other vehicle sensor may provide a signal indicative of a trailer being connected as represented at **624**. Similarly, the controller may detect continuity or a load from an electric circuit connected to the trailer to power trailer lights, etc. as represented at **626**. The controller may use one or more algorithms based on signals from various sensors to detect trailer towing or other significant increase in vehicle load as generally represented by performance algorithm **628**.

(33) If a trailer or other high demand use is detected at **620**, then the controller determines whether the SOC is within the battery health zone defined by first threshold (T1) and second threshold (T2) at **630**. If the SOC is within the health zone as determined at **630**, the controller operates the powertrain to sustain the SOC as represented at **640**. The controller may control the powertrain to a target SOC as represented at **650**. As previously described, charge sustaining operation may use a secondary power source, such as an internal combustion engine, a second battery, a fuel cell, etc. to provide propulsive torque directly to the vehicle wheels, or may power the electric machine to provide propulsive torque. The secondary power source may also charge the battery to maintain the SOC within a specified control range of the target SOC. Regenerative braking may also be used with the electric machine operating as a generator to charge the traction battery. Similarly, the traction battery may be discharged to provide additional acceleration or to maintain the target SOC while operating in the charge sustaining zone.

(34) If the SOC is not within the battery health zone as determined at **630**, then the controller determines if the SOC is below a third threshold (T3) at **660**. If yes at **660**, the controller operates the powertrain to sustain the SOC as indicated at **640** and sets a target SOC at **650** as previously described. If no at **660**, the controller operates the powertrain in a charge depletion zone or mode at **670** and continues to discharge the traction battery to power the electric machine to propel the vehicle.

(35) If a trailer or other high demand use is not detected at **620**, then the controller determines whether the SOC is below the third threshold (T3) at **660**. If yes, the controller operates the powertrain to sustain the SOC at **640** based on a corresponding target SOC indicated at **650**. If no at **660**, the controller operates the powertrain in a charge depletion mode or zone at **670**.

(36) As previously described, existing strategies for electrified vehicle control typically provide a single charge depletion zone and a single charge sustaining zone. As such, high demand applications such as trailer towing with associated high Ah throughput may adversely impact battery SoH. In contrast, one or more embodiments of the disclosure as described above utilize a battery health zone to increase operation in charge sustaining mode and reduce high Ah throughput over the vehicle lifetime to reduce or eliminate the impact on battery SoH.

(37) Use of the control strategy to increase operation of charge sustaining mode during trailer towing and similar high demand use scenarios may substantially improve battery life as demonstrated by an empirical analysis for a representative PHEV and summarized in the following table. The table illustrates Ah/mile over the vehicle life for a prior art baseline strategy with 20% trailer towing (TT) miles relative to the health zone strategy described herein:

(38) TABLE-US-00001 Baseline Health Zone Non-TT 1.33 1.33 TT 2.19 0.96 TT Usage 20% 20% Composite 1.50 1.26

In the table above, the composite baseline Ah/mile is 1.5 for a vehicle towing 20% of the miles. However, by implementing a battery health zone control strategy according to the disclosure, composite Ah throughput is reduced to 1.26 Ah/mile, which is a significant improvement of 16%. This improvement will increase even more for applications that have trailer towing or other high demand use scenarios as a larger portion of overall vehicle use, i.e. as high demand mileage increases over the 20% use case illustrated in the table.

(39) While representative embodiments are described above, it is not intended that these

embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the claimed subject matter. As previously described, the features of various embodiments can be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments may have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes may include, but are not limited to cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and can be desirable for particular applications.

Claims

1. An electrified vehicle comprising: a traction battery; an electric machine powered by the traction battery to selectively provide propulsion to the vehicle; an engine coupled to the electric machine; and a controller programmed to, in response to detecting a trailer connected to the electrified vehicle, operate the electric machine and the engine to sustain a state of charge (SOC) of the traction battery when the SOC is between a first threshold and a second threshold and when the SOC is below a third threshold, and operate the electric machine and the engine to deplete the SOC when the SOC is above the first threshold and when the SOC is between the second threshold and the third threshold.
2. The electrified vehicle of claim 1 wherein the first threshold is higher than the second threshold and the second threshold is higher than the third threshold.
3. The electrified vehicle of claim 1 further comprising at least one sensor mounted on the vehicle and configured to provide a signal to the controller to detect the trailer.
4. The electrified vehicle of claim 3 wherein the at least one sensor comprises a camera.
5. The electrified vehicle of claim 1 wherein the controller is programmed to detect the trailer when a trailer electrical circuit configured to power trailer lights is connected to a corresponding electrical circuit of the electrified vehicle.
6. The electrified vehicle of claim 1 wherein the controller is programmed to detect the trailer by comparing a measured vehicle acceleration to an expected vehicle acceleration associated with a driver demand torque.
7. The electrified vehicle of claim 1 further comprising a human-machine interface (HMI), wherein the controller is programmed to detect the trailer in response to input received by the HMI and communicated to the controller.
8. The electrified vehicle of claim 1 wherein the first threshold corresponds to 70% SOC and the second threshold corresponds to 30% SOC.
9. A method for controlling an electrified vehicle having a traction battery coupled to an electric machine to selectively provide propulsive torque to the vehicle, and a secondary power source configured to supply power to at least one of the electric machine and the traction battery, the method comprising, by a vehicle controller: in response to detecting an operating condition associated with traction battery amp-hour throughput exceeding an associated throughput threshold, controlling the electric machine and the secondary power source to sustain a state of charge (SOC) of the traction battery when the SOC is below a first threshold and above a second threshold, and when the SOC is below a third threshold; and in response to not detecting the operating condition, controlling the electric machine and the secondary power source to deplete the

SOC of the traction battery when the SOC is above the third threshold, and to sustain the SOC when the SOC is below the third threshold; wherein the first threshold is higher than the second threshold and the second threshold is higher than the third threshold.

10. The method of claim 9 further comprising, by the vehicle controller: in response to detecting the operating condition, controlling the electric machine and the secondary power source to deplete the SOC of the traction battery when the SOC is above the first threshold, and when the SOC is below the second threshold and above the third threshold.

11. The method of claim 10 wherein the secondary power source comprises an internal combustion engine.

12. The method of claim 11 wherein the operating condition comprises towing a trailer.

13. The method of claim 12 further comprising detecting the operating condition in response to receiving input from a human-machine interface (HMI) of the electrified vehicle.

14. The method of claim 12 further comprising detecting the operating condition in response to detection of completion of an electrical circuit between the trailer and the electrified vehicle.

15. The method of claim 12 further comprising detecting the operating condition based on comparing measured acceleration of the electrified vehicle for a specified torque request to an associated expected acceleration.

16. A system for an electrified vehicle having an electric machine powered by a traction battery and an engine, the system comprising: a controller programmed to, in response to detecting an operating condition associated with traction battery amp-hour throughput exceeding an associated throughput threshold: sustain a state of charge (SOC) of the traction battery when the SOC is either: a) below a first threshold and above a second threshold; or b) below a third threshold; and deplete the SOC of the traction battery when the SOC is between the second and the third thresholds.

17. The system of claim 16 wherein the controller is further programmed to detect the operating condition in response to detecting a trailer coupled to the electrified vehicle.

18. The system of claim 17, wherein the controller is further programmed to, in response to not detecting the operating condition: deplete the SOC when the SOC is above the third threshold; and sustain the SOC when the SOC is below the third threshold.

19. The system of claim 18 wherein the controller is further programmed to detect the operating condition based on input from a human-machine interface of the electrified vehicle.

20. The system of claim 19 wherein the controller is further programmed to detect the operating condition based on an electrical connection between the trailer and the electrified vehicle.